This week

- Microcontroller architecture contd.
- Microcontroller assembly programming

Terminology Test

- Microprocessor vs Microcontroller
- Harvard vs van Neuman architecutre

Harvard and von Neumann Architectures

- Harvard Architecture—a type of computer architecture where the instructions (program code) and data are stored in separate memory spaces
  - Example: Intel 8051 architecture

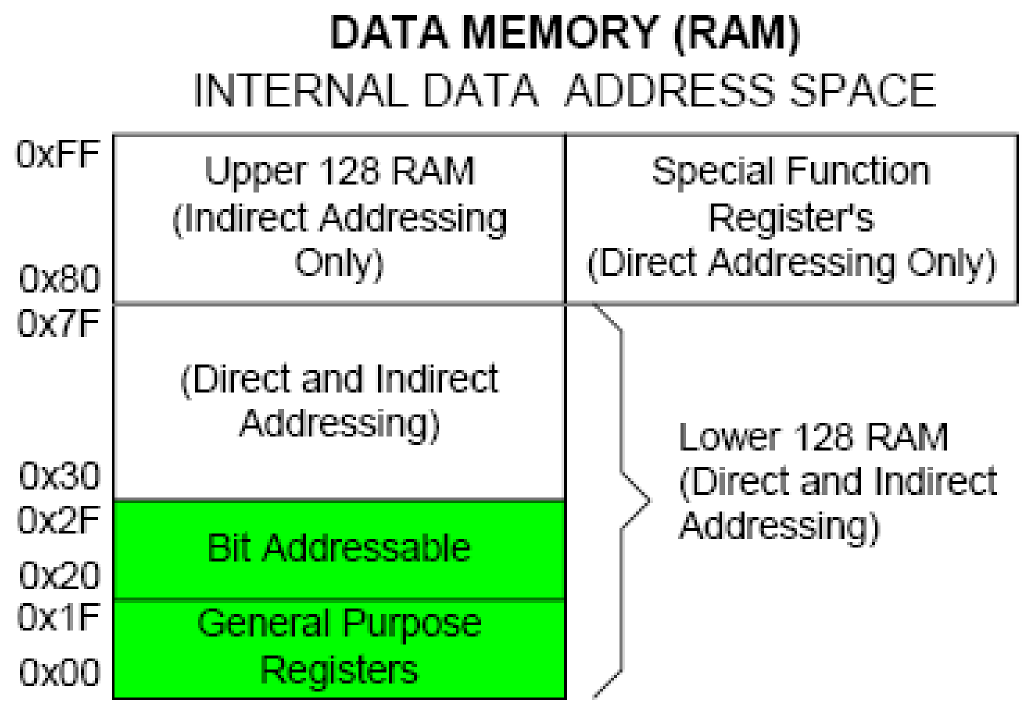
- von Neumann Architecture—another type of computer architecture where the instructions and data are stored in the same memory space
  - Example: Intel x86 architecture (Intel Pentium, AMD Athlon, etc.)

Data Memory (RAM)

- Internal Data Memory space is divided into three sections
  - Lower 128
  - Upper 128
  - Special function register (SFR)

- There are 384 bytes of memory space physically, though the Upper 128 and SFRs share the same addresses from location 80H to FFH.

- Appropriate instructions should be used to access each memory block



Lower 128—Register Banks and RAM

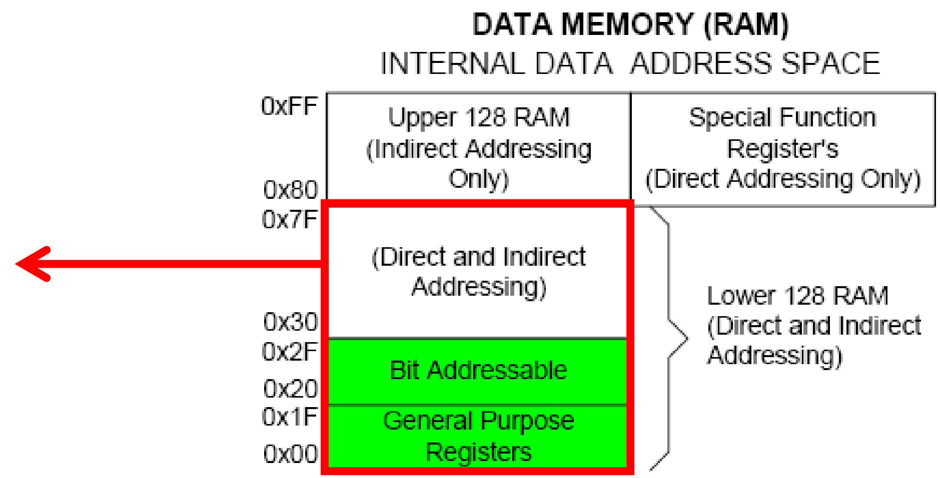
Byte Address	Bit Address							
7F	General Purpose RAM							
30								
2F	7F	7E	7D	7C	7B	7A	79	78
2E	77	76	75	74	73	72	71	70
2D	6F	6E	6D	6C	6B	6A	69	68
2C	67	66	65	64	63	62	61	60
2B	5F	5E	5D	5C	5B	5A	59	58
2A	57	56	55	54	53	52	51	50
29	4F	4E	4D	4C	4B	4A	49	48
28	47	46	45	44	43	42	41	40
27	3F	3E	3D	3C	3B	3A	39	38
26	37	36	35	34	33	32	31	30
25	2F	2E	2D	2C	2B	2A	29	28
24	27	26	25	24	23	22	21	20
23	1F	1E	1D	1C	1B	1A	19	18
22	17	16	15	14	13	12	11	10
21	0F	0E	0D	0C	0B	0A	09	08
20	07	06	05	04	03	02	01	00
1F	Bank 3							
18								
17	Bank 2							
10								
0F	Bank 1							
08								
07	Default Register Bank for R0 – R7							
00								

Bit Addressable

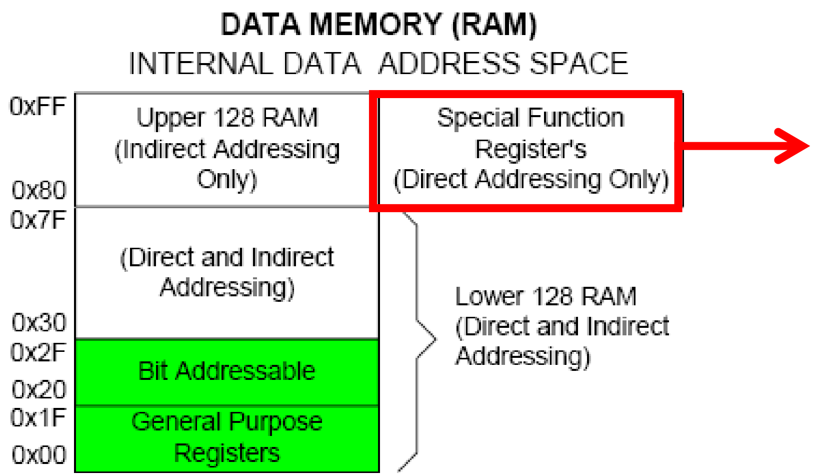
General Purpose RAM (80 bytes)

Bit-addressable Area (16 bytes)

Register Banks (8 bytes per bank; 4 banks)



Special Function Registers (SFRs)



- SFRs provide control and data exchange with the microcontroller's resources and peripherals
- Registers which have their byte addresses ending with 0H or 8H are *byte*- as well as *bit*- addressable
- Some registers are not bit-addressable. These include the stack pointer (SP) and data pointer register (DPTR)

Byte Address	Bit Address								
FF									
F0	F7	F6	F5	F4	F3	F2	F1	F0	B
E0	E7	E6	E5	E4	E3	E2	E1	E0	ACC
D0	D7	D6	D5	D4	D3	D2	-	D0	PSW
B8	-	-	-	BC	BB	BA	B9	B8	IP
B0	B7	B6	B5	B4	B3	B2	B1	B0	P3
A8	AF	-	-	AC	AB	AA	A9	A8	IE
A0	A7	A6	A5	A4	A3	A2	A1	A0	P2
99	Not bit-addressable								SBUF
98	9F	9E	9D	9C	9B	9A	99	98	SCON
90	97	96	95	94	93	92	91	90	P1
8D	Not bit-addressable								TH1
8C	Not bit-addressable								TH0
8B	Not bit-addressable								TL1
8A	Not bit-addressable								TL0
89	Not bit-addressable								TMOD
88	8F	8E	8D	8C	8B	8A	89	88	TCON
87	Not bit-addressable								PCON
83	Not bit-addressable								DPH
82	Not bit-addressable								DPL
81	Not bit-addressable								SP
80	87	86	85	84	83	82	81	80	P0

Summary of SFRs

- **Accumulator (ACC) and B register**
  - **ACC** (also referred to as **A**) is used implicitly by several instructions
  - **B** is used implicitly in multiply and divide operations
  - These registers are the input/output of the arithmetic and logic unit (ALU)
- **Program status word—PSW**
  - Shows the status of arithmetic and logical operations using multiple bits such as Carry
  - Selects the Register Bank (Bank 0 - Bank 3)
- **Stack pointer—SP**
- **Data pointer—DPTR (DPH and DPL)**
  - 16-bit register used to access external code or data memory
- **Timer Registers—TH0, TL0, TH1, TL1, TMOD, TCON**
  - Used for timing intervals or counting events
- **Parallel I/O Port Registers—P0, P1, P2** and **P3**
- **Serial Communication Registers—SBUF** and **SCON**
- **Interrupt Management Registers—IP** and **IE**
- **Power Control Register—PCON**

PSW: PROGRAM STATUS WORD. BIT ADDRESSABLE.

CY	AC	F0	RS1	RS0	OV	—	P

CY	PSW.7	Carry Flag.
AC	PSW.6	Auxiliary Carry Flag.
F0	PSW.5	Flag 0 available to the user for general purpose.
RS1	PSW.4	Register Bank selector bit 1 (SEE NOTE 1).
RS0	PSW.3	Register Bank selector bit 0 (SEE NOTE 1).
OV	PSW.2	Overflow Flag.
—	PSW.1	User definable flag.
P	PSW.0	Parity flag. Set/cleared by hardware each instruction cycle to indicate an odd/even number of '1' bits in the accumulator.

NOTE:
1. The value presented by RS0 and RS1 selects the corresponding register bank.

RS1	RS0	Register Bank	Address
0	0	0	00H-07H
0	1	1	08H-0FH
1	0	2	10H-17H
1	1	3	18H-1FH

Whats next !

- Lets use it in more programming exercises !!!